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XI Jornadas Internacionales de Investigación en Ciberseguridad

Formalization and analysis of the post-quantum signature scheme FALCON with Maude

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1. Motivation
2. FALCON
3. Framework
4. Experiments
 - LTL Model Checking
 - Invariant Analysis
5. Conclusions & Future Work

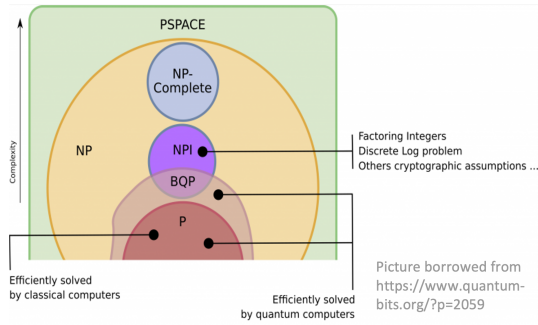
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Threat of adversaries with quantum capabilities

- Shor's algorithm to solve discrete logarithm problem
- Grover's search algorithm

Search of solutions by NIST with the PQC project (round 3)

- Key Encapsulation Mechanisms: CRYSTALS-Kyber
- Digital Signature Schemes: CRYSTALS-Dilithium, **FALCON**, SPHINCS+



What guarantees do we have on these new mechanisms?

Two main approaches for security analysis exist:

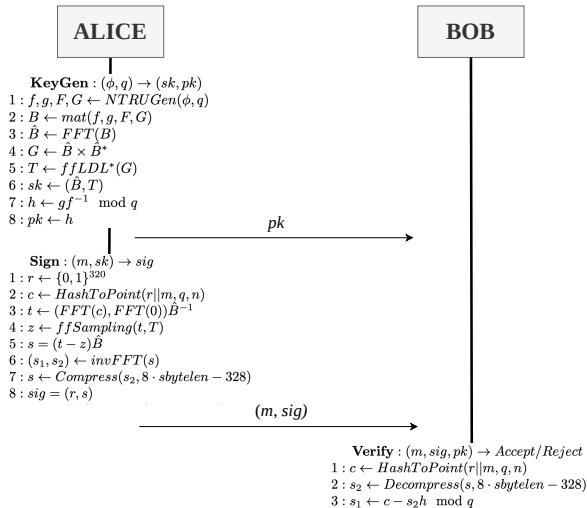
Computational:

- Use of mathematical proofs and probabilities
- Keys, messages,... are bit strings
- Closer to reality, used by cryptographers

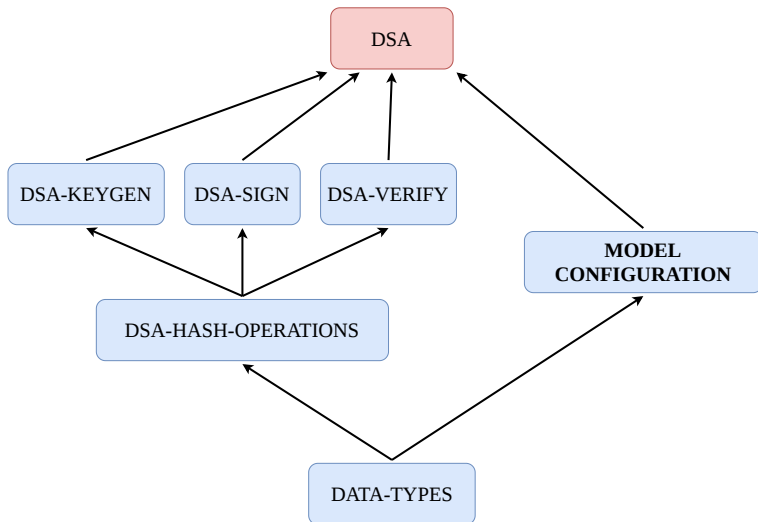
Symbolic:

- Cryptographic primitives as black boxes
- Keys, messages,... are symbols
- Focuses on the properties of the theory underneath
- Suitable for automation
- Easier to understand for non-experts of cryptography

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Termination: eventual application of the verification algorithm.

$$\diamond \text{verifiedSignatureFrom}(ID1, ID2)$$

Fairness: whenever the sender wants to send a signed message to the receiver, the receiver will eventually verify it.

$$\square (\text{wantsToSignAndSendTo}(ID1, ID2)$$

$$\rightarrow$$

$$\diamond \text{verifiedSignatureFrom}(ID2, ID1))$$

BOTH HOLD!

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Authentication: there is a proof of the sender having the private key and signing the message with it. In other words, the receiver of the signature should not be able to verify a message if it has not been signed by the sender first.

Proof: we search for paths in which a message has been verified without it being signed at first by the other participant.

Result: MITM attack due to no security guarantees/assumptions on the distribution of the public key.

Integrity: proof of a message not being modified after the signature generation process.

Proof: we search for paths in which a message verified by the receiver is **NOT** the same that the sender generated and signed.

Result: No solution, thus property holds.

Non-repudiation: proof of a participant not being able to deny the signature for a message he has signed.

Proof: we search for paths in which a signature for a message not signed by the sender **CAN** be verified by the receiver.

Result: No solution, thus property holds.

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Conclusions

- Extended the framework to handle DSAs.
- Specified an executable symbolic model of FALCON.
- Checked termination and fairness of our model with LTL Model Checking.
- Performed invariant analysis for Authentication, Integrity and Non-Repudiation.

Future Work

- Consider different types of attackers.
- Replicate known attacks (e.g. Peregrine).
- Keep improving the framework to use object notation.
- Use of advanced techniques to cover unbounded number of participants and attackers.